

LAB 3 – COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Problem Statement: Household Carbon Footprint & Target Tracker System

1. Scenario Review

The system allows a Resident to record monthly electricity, gas and commute consumption, calculate household CO₂e emissions, view historical emissions, set a reduction milestone and track progress. A Sustainability Coach can review a Resident's emissions history and target progress.

2. Requirements Considered

Type	ID	Requirement
Functional	FR-001	Calculate total monthly CO ₂ e from electricity and transport data.
Functional	FR-002	Record monthly electricity, gas and commute consumption.
Functional	FR-003	Display monthly CO ₂ e emissions and historical trend.
Functional	FR-004	Set a CO ₂ e reduction milestone and show progress.
Functional	FR-005	Allow the Sustainability Coach to review emissions history and target progress.
Non-functional	NFR-001	Annual reduction milestone chart renders in under 200 ms.
Non-functional	NFR-002	Only the authorized Resident and assigned Sustainability Coach can access household data.

3. Architectural Style Analysis

Architecture	Advantages	Limitations	Decision
Layered	Clear separation of presentation, business logic and data; simple to understand and maintain.	Some communication overhead between layers; less independent scaling.	Selected
Microservices	Independent deployment, scaling and fault isolation.	Adds operational complexity, network communication and data-consistency issues.	Not selected for this scope
Client-Server	Centralised control and straightforward data consistency.	Central server can become a bottleneck and a single point of failure.	Not selected

4. Architecture Selection

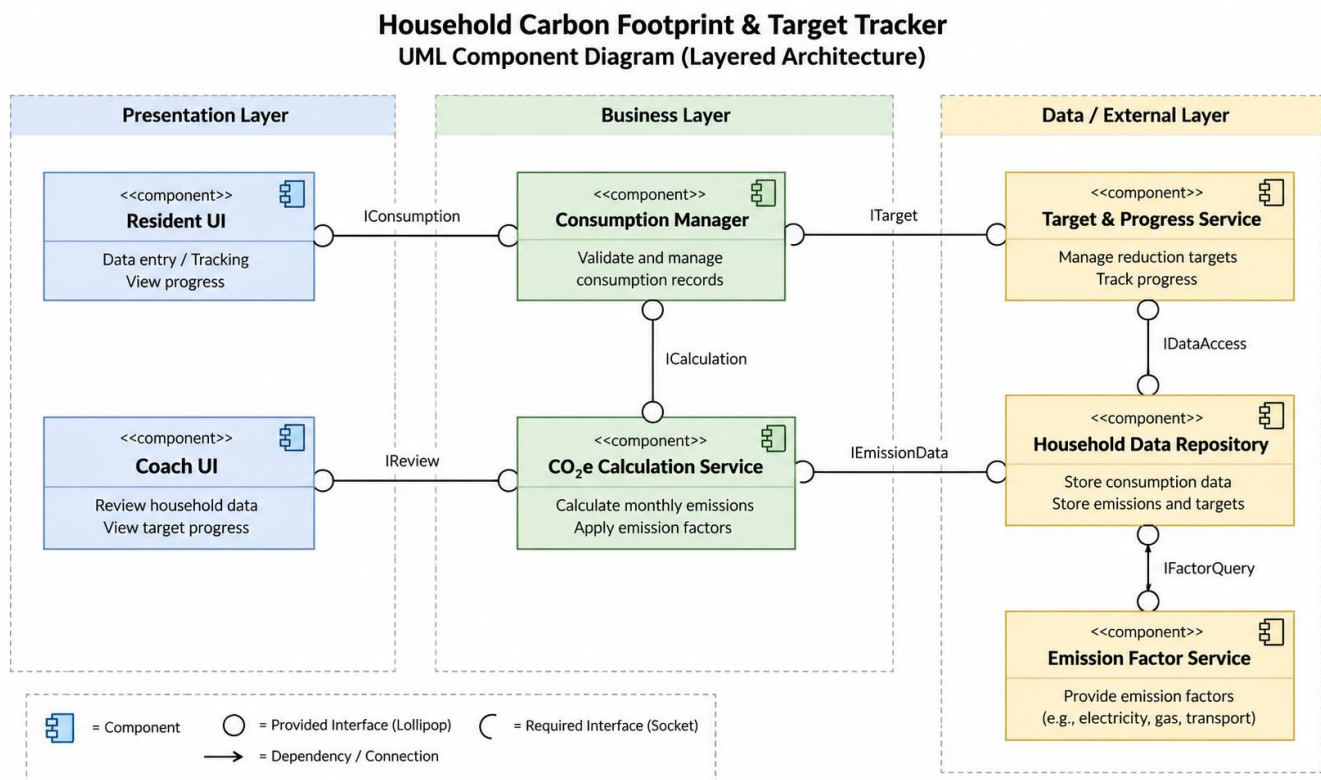
Layered Architecture was selected because the system naturally separates into user-facing functions, business processing and persistent data. It gives a practical structure for the current system without introducing the operational complexity of microservices.

5. Components

Component	Responsibility
Resident UI	Resident data entry, emissions and target screens.
Coach UI	Sustainability Coach review screen.
Consumption Manager	Validates and manages monthly consumption records.
CO ₂ e Calculation Service	Calculates monthly emissions using consumption data and emission factors.
Target & Progress Service	Manages reduction milestones and progress.
Household Data Repository	Stores consumption, emissions and target information.
Emission Factor Service	External service providing conversion/emission factors.

6. Component Diagram

The diagram contains seven components and the main interfaces between them.



7. Interfaces and Interactions

Interface	Direction	Purpose
Consumption API	Resident UI → Consumption Manager	Monthly electricity, gas and commute data.
Target API	Resident UI → Target & Progress Service	Target and progress requests.
Review API	Coach UI → Target & Progress Service	Review household progress.
Calculation Request	Consumption Manager → CO ₂ e Calculation Service	Request monthly CO ₂ e calculation.
Emission Record	CO ₂ e Calculation Service → Data Repository	Store calculated emissions.
Target Data	Target & Progress Service → Data Repository	Read/write target and progress data.
Factor Query	CO ₂ e Calculation Service → Emission Factor Service	Obtain emission factors.

8. Security and Performance

Security is addressed by keeping household data behind application components and applying authorization before protected records are returned to the Resident or assigned Sustainability Coach. Performance is supported by separating calculation and data-access responsibilities, helping the system meet the under-200-ms chart-rendering requirement.